

How to run my research again by myself

Step-by-step guide for papers 1, 2 and 3

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This is a practical guide. It lists, in order, every command needed to rebuild all the results of Reports 1–3 on my MacBook, starting from nothing but the code. After each step there is a **Check** with the number I should see, so I know at once if something went wrong. The physics behind each step is in Reports 1–3 (the files `NEUTRON_STARS_DOC-2.pdf` and `-3.pdf` are Reports 2 and 3; their content is the same as `paper2_neutron_skin.pdf` and `paper3_magnetized_envelope.pdf`).

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1 The whole chain in one picture

Step	What	Program	Output
1	EoS of the core, for every ($E_{\text{sym}}, L_{\text{sym}}$)	rmf_lambda_and_eos.py	EOS_RMf/eos<TAG>.dat
2	Crust join + NSCool format	automationandformatting-2.py, fix_seam_eos.py	NSCool/EOS/<TAG>.CAT.dat
3	TOV: $M-R$ curve and star profiles	toVprofile (Fortran)	MRcurve.<TAG>_CAT.dat, TOVprofile.*
4	Keep EoS with $M_{\text{max}} \geq 2$ and causal	export_cooling_chi2.py	Data/surviving-tags.txt (864)
5	Cooling curves	run_cooling_prl.py + NSCool	Data/Cooling_Outputs/...
6	χ^2 against 16 stars	export_final_chi2.py	Data/cooling_chi2_final.csv
7	Intervals, M_{DU} , figures	physical_quantities.py, combine_cooling_lab.py, ...	numbers for paper 1
8	Paper 2: skins of ^{208}Pb , ^{48}Ca	finite_nucleus_rmf.py, rerun_skin_grid_fixed.py	skin_grid*.csv
9	Paper 3: magnetised envelopes	run_cooling_prl.py (magB modes)	Data/cooling_chi2_magnetized.csv
10	Verification and LaTeX	Verification/*.py, pdflatex	PDFs

A *tag* is the name of one EoS, for example GM2_J32p5_L80p0 means model GM2 with $E_{\text{sym}} = 32.5$ and $L_{\text{sym}} = 80.0$ MeV (the dot is written p).

2 Step 0: set up the Mac (once)

2.1 Programs

```
xcode-select --install          # compilers, git, make
brew install gcc                # gives gfortran (needs Homebrew, brew.sh)
python3 -m pip install numpy scipy matplotlib
```

For LaTeX install MacTeX (<https://tug.org/mactex/>).

Check: `gfortran --version` and `python3 -c "import numpy, scipy, matplotlib"` both work without error.

2.2 Folder layout

All paths below are relative to my main folder `~/phdfolder`. The scripts expect this layout (they find each other through it):

```
phdfolder/
  Source_Code/
    run_cooling_prl.py  make_workers.py      <- cooling drivers
    Python/             rmf_lambda_and_eos.py, automationandformatting-2.py,
                       fix_seam_eos.py, audit_eos_tables.py,
                       finite_nucleus_rmf.py, skin_grid*.csv, ...
  NSCool/              EOS/  TOV/  I_Files/  Code/  Model_1/  (NSCool itself)
  Data/                Cooling_Outputs/, cooling_chi2_final.csv, ...
  Plotting_Scripts/    export_final_chi2.py, joint_likelihood.py, ...
  Verification/        verify_papers.py, validate_solver.py, ...
  Paper/PRC_papers/    the paper .tex files and figs/
```

Careful: `run_cooling_prl.py` and `make_workers.py` look for NSCool/ one folder above their own folder, while the scripts in Source_Code/Python/ look two folders up. If a script says it cannot find NSCool/EOS, the script is in the wrong folder. Check with `ls Source_Code Source_Code/Python`.

2.3 Get the corrected files from GitHub

All the corrections of the last weeks are on GitHub, branch `claude/focused-fermi-15kqdj`:

```
cd ~
git clone -b claude/focused-fermi-15kqdj \
  https://github.com/nishasinghkuntal/phdfolder.git phd_updates
```

Then copy into the working tree (first make a backup of the old versions):

```
cd ~/phdfolder
P=Source_Code/Python
mkdir -p backup_old && cp $P/finite_nucleus_rmf.py $P/skin_grid*.csv backup_old/
U~/phd_updates
cp $U/finite_nucleus_rmf.py $U/rerun_skin_grid_fixed.py $U/durca_threshold_mass.py \
  $U/physical_quantities.py $U/combine_cooling_lab.py $P/
mkdir -p $P/reports/data && cp $U/reports/data/*.dat $P/reports/data/
cp $U/skin_grid_fixed*.csv $U/skin_validation_fixed.csv $P/
cp $U/papers/paper1_cooling.tex $U/papers/paper2_skins.tex Paper/PRC_papers/
cp -r $U/reports ~/phdfolder/Reports
```

Later, to pick up new changes: `cd ~/phd_updates && git pull` and copy again.

2.4 NSCool

Compile NSCool as before (in NSCool/Code, with its own Makefile). My copy carries the `bfr` patch (the magnetic field was never passed to the envelope; see Report 3). After compiling:

```
python3 Verification/verify_bfield_patch.py
```

Check: it ends with “Both assertions hold: the bug is characterised and the patch is ...”. If it prints FAILED, the patch is missing and paper 3 results cannot be reproduced.

3 Step 1: the equations of state

```
cd ~/phdfolder/Source_Code/Python
python3 rmf_lambda_and_eos.py          # full grid + accuracy checks
python3 rmf_lambda_and_eos.py --checks # only the accuracy checks (fast)
```

What it does: for each model (GM1, GM2, FSUGarnet, IOPB-I, BigApple) it keeps the published isoscalar couplings, and for every grid point ($E_{\text{sym}} = 25\text{--}43$ MeV in steps of 1.5, $L_{\text{sym}} = 30\text{--}130$ MeV in steps of 5, plus the extra GM points) it solves for g_ρ and $\Lambda_{\omega\rho}$, then solves β -equilibrium from 0.03 to about 1.5 fm^{-3} . It writes EOS_RMF/eos<TAG>.dat and couplings_table.dat. Points where $\Lambda_{\omega\rho}$ or C_ρ^2 come out unphysical are skipped and printed.

Check: GM1 at $E_{\text{sym}} = 32.5$, $\Lambda_{\omega\rho} = 0$: $C_\rho^2 = 4.414 \text{ fm}^2$ (published 4.411); GM2/GM3: 4.794 (4.791). NL3 $\omega\rho$ test: $\Lambda_{\omega\rho} = 0.0297$.

Check: `grep -c . couplings_table.dat` gives about 1478 lines (all tables before cuts).

Careful: The code must pass the *Landau* effective mass $\sqrt{k_F^2 + M^{*2}}$ to NSCool, not the Dirac mass. Check once: `grep -n "MASS_FOR_NSCOOL" rmf_lambda_and_eos.py` must show "landau".

4 Step 2: crust and NSCool format

```
python3 automationandformatting-2.py EOS_RMF/eos*.dat
python3 fix_seam_eos.py          # dry run: lists tables with a bad seam
python3 fix_seam_eos.py --apply  # repairs them
python3 audit_eos_tables.py      # checks every *_CAT.dat
```

The core is joined to the HZD-NV crust of NSCool at $n = 0.0789 \text{ fm}^{-3}$ ($Z = 32$, $A = 982$), where the pressure matches within 5%. Output: NSCool/EOS/<TAG>.CAT.dat.

Check: `fix_seam_eos.py` lists 11 tables before `--apply`, and 0 after. `audit_eos_tables.py` reports no pressure inversions and no $c_s > c$ in the crust.

Careful: Do not write your own parser for the `_CAT.dat` files. They contain several blocks with different columns; a simple “read all numbers” parser invents fake pressure inversions. This happened twice. Use `audit_eos_tables.py` or `Verification/verify_eos_thermo.py`.

5 Step 3: TOV

```
cd ~/phdfolder
gfortran -O2 -o toVprofile Source_Code/Python/TOVprofile2_automated.f90
./toVprofile NSCool/EOS/GM1*_CAT.dat          # one model at a time
./toVprofile NSCool/EOS/GM2*_CAT.dat
./toVprofile NSCool/EOS/FSUGarnet*_CAT.dat
./toVprofile NSCool/EOS/IOPB-I*_CAT.dat
./toVprofile NSCool/EOS/BigApple*_CAT.dat
```

(Use the path where `TOVprofile2_automated.f90` actually is.) The program must be started from `phdfolder` (it looks for `NSCool/EOS/APR.EOS_Cat.dat`). It accepts at most 1024 files per call, so run it model by model. For a quick M – R curve only, add `--curve-only`.

Output: `NSCool/EOS/MRcurve_<TAG>_CAT.dat` (columns M , R , ρ_c) and the star profiles `NSCool/TOV/Profile/TOVprofile_<TAG>_targetM10.dat`, ... `M14`, `M18`, `M20` that NSCool needs.

Check: with $\Lambda_{\omega\rho} = 0$, $E_{\text{sym}} = 32.5$: $M_{\text{max}}(\text{GM1}) = 2.360$, $\text{GM2} = 2.075$, $\text{GM3} = 2.015 M_{\odot}$ (published 2.35–2.36, 2.08, 2.02). At the published $(E_{\text{sym}}, L_{\text{sym}})$: FSUGarnet 2.073, IOPB-I 2.120, BigApple 2.593 $M_{\odot}$ (Report 1, Table 2).

Check: independent check with the Python TOV: `python3 Source_Code/Python/durca_threshold_mass.py` prints `GM1  $M_{\text{max}} = 2.361$` .

6 Step 4: which EoS survive

The cut is $M_{\text{max}} \geq 2.0 M_{\odot}$, causal ($c_s < c$ everywhere in the stable star), $E_{\text{sym}} \in [25, 43]$, $L_{\text{sym}} \in [30, 130]$. The plotting scripts already contain it; I just write the list of tags to a file:

```
cd ~/phdfolder/Plotting_Scripts
python3 -c "import export_cooling_chi2 as X; e=X.build_eos_data(); \
t=sorted(k for k,v in e.items() if v['pass_all']); \
open('../Data/surviving_tags.txt', 'w').write('\n'.join(t)+'\n'); print(len(t))"
```

Check: it prints **864**. By model: GM1, GM2 with the extra GM points, and FSUGarnet, IOPB-I, BigApple on the regular grid.

7 Step 5: cooling with NSCool

7.1 Parallel workers

NSCool writes its temporary files into `Model.1/` with fixed names, so two runs in the same folder overwrite each other silently (this corrupted a batch once). Always use separate workers:

```
cd ~/phdfolder
python3 Source_Code/make_workers.py 4          # makes NSCool_w1 ... NSCool_w4
# split the 864 tags into 4 lists of 216:
awk '{print > ("Data/tags_part_0" (NR%4))}' Data/surviving_tags.txt
```

7.2 The runs for paper 1

Paper 1 uses two modes: `durca_sf_vc` (direct Urca + modified Urca + superfluidity + vortex-creep heating, iron envelope) and `durca_sf_vc_carbon` (same, carbon envelope with $\eta = 6 \times 10^{-10}$, used only for Cas A and XMMU J1732), at four masses. Test first:

```
python3 Source_Code/run_cooling_prl.py --modes durca_sf_vc --masses 1.4 \
--max 5 --workdir NSCool_w1
```

Check: 5 runs finish with status ok; `Data/Cooling_Outputs/durca_sf_vc_M14/<TAG>/Teff_<TAG>.dat` exists. Note the time per run: the full job is $864 \text{ EoS} \times 4 \text{ masses} \times 2 \text{ modes} = 6912$ runs.

Then start the four workers, each in its own Terminal window (or with `nohup ... &`):

```
for i in 0 1 2 3; do
  nohup python3 Source_Code/run_cooling_prl.py \
    --modes durca_sf_vc durca_sf_vc_carbon --masses 1.0 1.4 1.8 2.0 \
    --tagfile Data/tags_part_0$i --workdir NSCool_w$((i+1)) \
    > Data/cool_w$((i+1)).log 2>&1 &
done
tail -f Data/cool_w1.log          # watch progress (Ctrl-C only stops tail)
```

Runs that already exist are skipped (`skip`), so if the Mac sleeps or crashes, start the same command again and it continues.

Careful: Keep the Mac awake: `caffeinate -i` in another window. A run that takes more than 300 s is killed and marked `fail`; a few failures at $2.0 M_{\odot}$ for soft EoS are normal, many are not.

Check: at the end each log shows `fail = 0` (or a handful) and `no_profile = 0`. `no_profile` means the TOV profile for that mass is missing: go back to Step 3.

8 Step 6: χ^2 against the observations

```
cd ~/phdfolder/Plotting_Scripts
python3 export_final_chi2.py
```

It compares every EoS with the 16 stars (list OBS in `full_multimass_5mode.py`), takes for each star the best of the four masses with a Gaussian mass prior $1.4 \pm 0.3 M_{\odot}$, does not extrapolate curves beyond their end, and writes `Data/cooling_chi2_final.csv`.

Check: EoS scored: 864; best fit GM2, $J = 32.5$, $L = 80$; $\chi^2/N = 1.397$, $\chi^2/\text{dof} = 1.596$.

Careful: RX J0822–4300 must be $\log T_{\text{eff}}^{\infty} = 6.072$ (whole surface), not the hot-spot value. Check: `grep -n "J0822" full_multimass_5mode.py`.

9 Step 7: intervals, M_{DU} and figures (paper 1)

9.1 The $E_{\text{sym}} - L_{\text{sym}}$ intervals

```
python3 joint_likelihood.py      # also gives the paper-2 joint numbers (Step 8)
python3 make_final_figures.py   # figures into Paper/PRC_papers/figs/
```

Check: $L_{\text{sym}} = 50\text{--}75 \text{ MeV}$ (68%), $35\text{--}80$ (95%); $E_{\text{sym}} = 28\text{--}36$ (68%), $25\text{--}40$ (95%); $R_{1.4} = 12.96\text{--}13.63 \text{ km}$.

9.2 Direct Urca threshold mass and the heavy-ion comparison

```
cd ~/phdfolder/Source_Code/Python
python3 physical_quantities.py --tagfile ../../Data/surviving_tags.txt # ~10 min
python3 combine_cooling_lab.py ../../Data/cooling_chi2_final.csv
```

The first writes `reports/data/eos_physical_quantities.dat` (K_{sym} , $S(2n_0)$, $x_p(2n_0)$, n_{DU} , M_{DU} , M_{max} , $R_{1.4}$ for each EoS). The second gives 68/95% intervals of these quantities for cooling alone, cooling + INDRA-FAZIA (Montefusco et al., arXiv:2609.24940), and INDRA-FAZIA alone, plus $P(M_{\text{DU}} \leq 1.4 M_{\odot})$.

Check: the file has 864 data lines. Examples: GM2 $J = 32.5$, $L = 80$ gives $M_{\text{DU}} \approx 1.47 M_{\odot}$. Inside the 68% box the median M_{DU} is about $1.77 M_{\odot}$. Proton fraction at threshold $x_{p,\text{DU}} \approx 0.137$.

Careful: these two numbers have not yet been computed with the *final* `cooling_chi2_final.csv` (I only had an older file). After running this, put $P(M_{\text{DU}} \leq 1.4)$ and the combined interval into paper 1, Sec. “Direct Urca threshold masses”.

10 Step 8: neutron skins (paper 2)

```
cd ~/phdfolder/Source_Code/Python
python3 finite_nucleus_rmf.py BigApple 31.32 39.80 # one test nucleus run
python3 ../../Verification/validate_solver.py # published-point check
python3 rerun_skin_grid_fixed.py # whole skin grid, 4 cores
```

The grid run takes several hours (1–4 min per ^{208}Pb point). It resumes where it stopped if interrupted. It writes `skin_grid_fixed_Pb208.csv`, `skin_grid_fixed_Ca48.csv` and `skin_validation_fixed.csv`.

Check: `skin_validation_fixed.csv`: $\Delta r_{np}(\text{Pb, Ca}) = \text{FSUGarnet } 0.1636, 0.1676$; `IOPB-I` 0.2212, 0.2010; `BigApple` 0.1494, 0.1683 fm. Benchmarks inside `finite_nucleus_rmf.py`: `NL3` Pb 0.277 fm (published 0.280), `FSUGold` 0.207 (0.207). All 74 grid points per nucleus converged.

Then make the fixed grid the one the analysis reads, and run the likelihood:

```
cp skin_grid_fixed_Pb208.csv skin_grid_Pb208.csv
cp skin_grid_fixed_Ca48.csv skin_grid_Ca48.csv
cd ../../Plotting_Scripts && python3 joint_likelihood.py
```

Check: `PREX-2` alone: $L_{\text{sym}} = 50\text{--}95$, $E_{\text{sym}} = 35.5\text{--}43$ MeV; `CREX` alone: $L_{\text{sym}} = 50\text{--}70$, $E_{\text{sym}} = 26.5\text{--}32.5$ MeV (68%). Joint best fit `FSUGarnet` $E_{\text{sym}} = 31$, $L_{\text{sym}} = 55$ with $\Delta r_{np} \approx 0.174$ (Pb), 0.173 fm (Ca), $\chi^2 = 4.46$.

Careful: `GM1` and `GM2` are never used for skins (no meson masses published). The binding energies come out 0.09–0.24 MeV/A too small; this is an open issue and does not change the skins.

11 Step 9: magnetised envelopes (paper 3)

Same as Step 5 with the field modes (iron envelope with $B = 10^{11}\text{--}10^{14}$ G, and `magB0` as control that must equal `durca_sf_vc`):

```
for i in 0 1 2 3; do
  nohup python3 Source_Code/run_cooling_prl.py \
    --modes durca_sf_vc_magB0 durca_sf_vc_magB11 durca_sf_vc_magB12 \
      durca_sf_vc_magB13 durca_sf_vc_magB14 \
    --masses 1.0 1.4 1.8 2.0 --tagfile Data/tags_part_0$i \
    --workdir NSCool_w$((i+1)) > Data/mag_w$((i+1)).log 2>&1 &
done
```

This is $5 \text{ modes} \times 3456 = 17280$ runs, so it is the longest step. Then:

```
cd Plotting_Scripts
python3 export_magnetized_chi2.py      # writes Data/cooling_chi2_magnetized.csv
python3 make_paper3_figures.py
python3 ../Verification/verify_paper3.py
```

Check: `magB0` curves are identical to `durca_sf_vc` (this tests the `bf_r` patch in a second way). The numbers to compare are in Report 3.

12 Step 10: verification and papers

```
cd ~/phdfolder
python3 Verification/verify_eos_thermo.py    # thermodynamics of all 864 tables
python3 Verification/verify_papers.py       # numbers in papers vs data files
python3 Verification/verify_paper3.py
python3 Verification/verify_documents.py     # no retired numbers in other documents
python3 Verification/check_dois.py          # needs internet (Crossref)
```

All must end without FAIL. If `verify_papers.py` fails after a rerun, the new number is probably right and the paper text is old: update the paper, not the check.

Compiling the papers.

```
cd Paper/PRC_papers
pdflatex paper1_cooling && bibtex paper1_cooling
pdflatex paper1_cooling && pdflatex paper1_cooling # same for paper2, paper3
python3 ../../Plotting_Scripts/to_mnras.py        # regenerates the MNRAS versions
python3 ../../Verification/verify_mnras.py
```

Careful: The MNRAS files are generated. Never edit them by hand: edit the REVTeX file and run `to_mnras.py` again.

If LaTeX stops with “no legal `\end` found”: `python3 Reports/check.tex.py` `paper1_cooling.tex` shows the missing `\end{document}`, unclosed environment or unbalanced brace.

13 When something goes wrong

Symptom	What to do
Cooling curves of two EoS look identical, or change between runs <code>no_profile</code> in the cooling log	Two jobs used the same NSCool folder. Delete that batch, always use <code>--workdir NSCool.wN</code> . TOV profile for that mass missing: rerun Step 3 for that model.
Many fail (timeout)	Check the <code>_CAT.dat</code> with <code>audit_eos_tables.py</code> ; seam problems make NSCool slow. Run <code>fix_seam_eos.py --apply</code> .
Number of surviving EoS $\neq 864$	A model’s MR curves are missing or old. Compare <code>ls NSCool/EOS/MRcurve_* wc -l</code> before and after Step 3.
χ^2 changed a lot	Check RX J0822 value (6.072), the mass prior 0.3, and that both modes finished for all masses.
Skin run stops or is slow	It resumes; just start it again. Check that 4 cores are free.
Script cannot find NSCool/EOS	Script is in the wrong folder (Sec. 2.2), or start the Fortran program from <code>phdfolder</code> .

14 Things still open before submission

1. Run Step 7b with the final `cooling_chi2_final.csv` and Step 8 `joint_likelihood.py` with the fixed skin grid, then fill the numbers marked `% CHECK` in `paper1_cooling.tex` and `paper2_skins.tex`.
2. Find the original references for RX J0822–4300 (Potekhin et al. 2020, Table 1), and optionally rerun with $\sigma = 0.12$ for this star.
3. Confirm the published ζ values of FSUGarnet, IOPB-I, BigApple; decide whether to rerun the scan with them (effect on $M_{\max} \leq 0.03 M_{\odot}$).
4. Binding-energy underbinding in the skin code (paper 2).
5. Ask D. Page about the `bf_r` bug in the public NSCool (paper 3).